Identification of gene expression signatures associated with cisplatin intrinsic resistance in lung adenocarcinoma cell lines

Chavez-Dominguez R[1,5], Perez-Medina M[2,5], Soto-Nava M[3], Avila-Rios S[3], Mendez-Tenorio A[4], Galicia-Velasco M[5], Lopez-Gonzalez J[5], Aguilar-Cazares D[5]

1. Posgrado en Ciencias Biologicas, Universidad Nacional Autonoma de Mexico, Ciudad de Mexico, Mexico.
2. Escuela Nacional de Ciencias Biologicas, Instituto Politecnico Nacional, Ciudad de Mexico, Mexico.
3. Instituto Nacional de Enfermedades Respiratorias, Centro de Investigacion de Enfermedades Infecciosas, C.P 14080, Ciudad de Mexico, Mexico.
4. Laboratorio de Biotecnologia y Bioinformatica Genomica, Departamento de Bioquimica, Escuela Nacional de Ciencias Biologicas, Instituto Politecnico Nacional, CP 11340, Ciudad de Mexico, Mexico
5. Instituto Nacional de Enfermedades Respiratorias, Departamento de Enfermedades Cronico-Degenerativas, Laboratorio de Cancer Pulmonar, C.P 14080, Ciudad de Mexico, Mexico

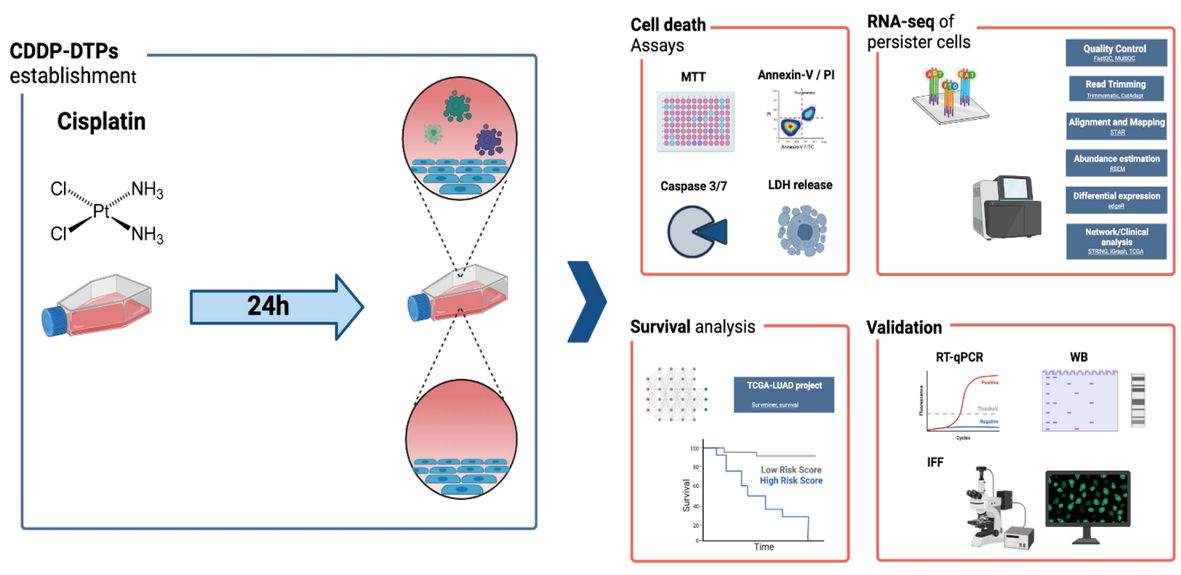
1. BACKGROUND

- ✓ Lung cancer is the leading cause of cancer-related mortality worldwide[1]
- ✓ CDDP is the standard care treatment for advanced lung cancer[2]
- ✓ Approximately, 60% of CDDP-treated patients do not response at treatment onset because of intrinsic resistance[3]
- ✓ **Drug-tolerant persister (DTPs) cells spare during drug treatment and are responsible for intrinsic and acquired resistance[4]**

2. OBJECTIVE

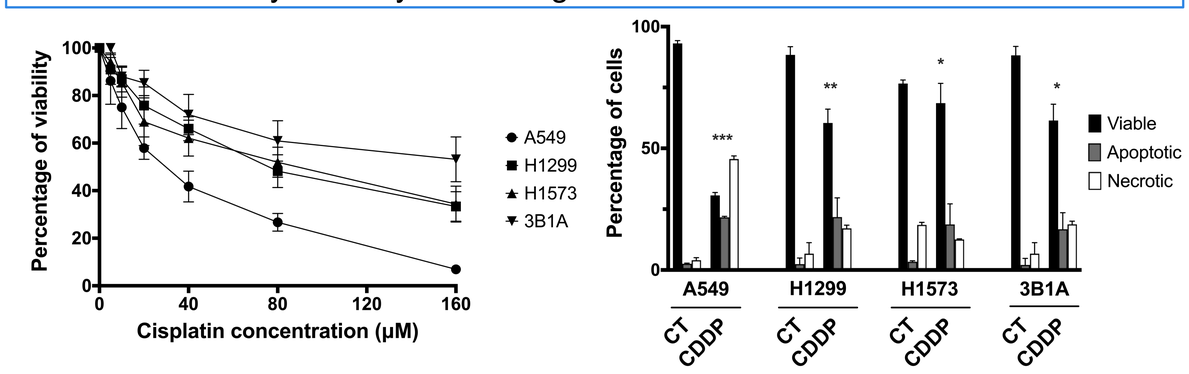
◎ Study the main molecular alterations at transcriptome level associated with the emergence of CDDP-drug-tolerant persister cells in lung adenocarcinoma cell lines

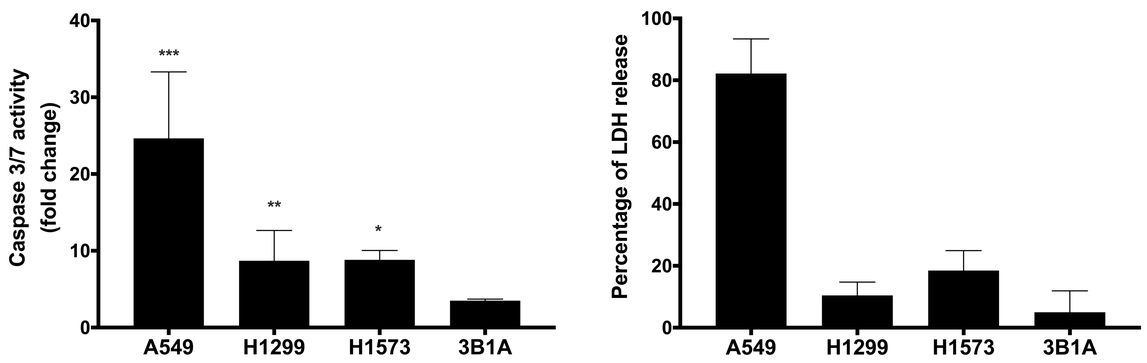
3. METHODS

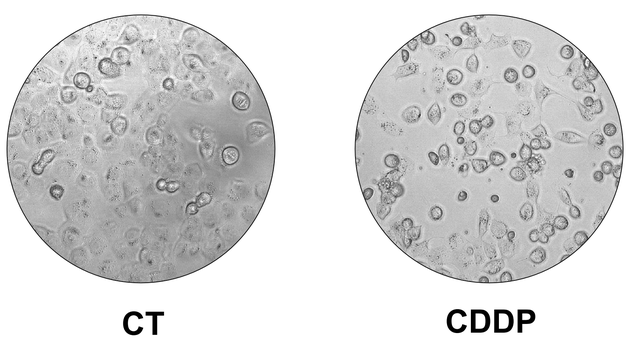


4. RESULTS

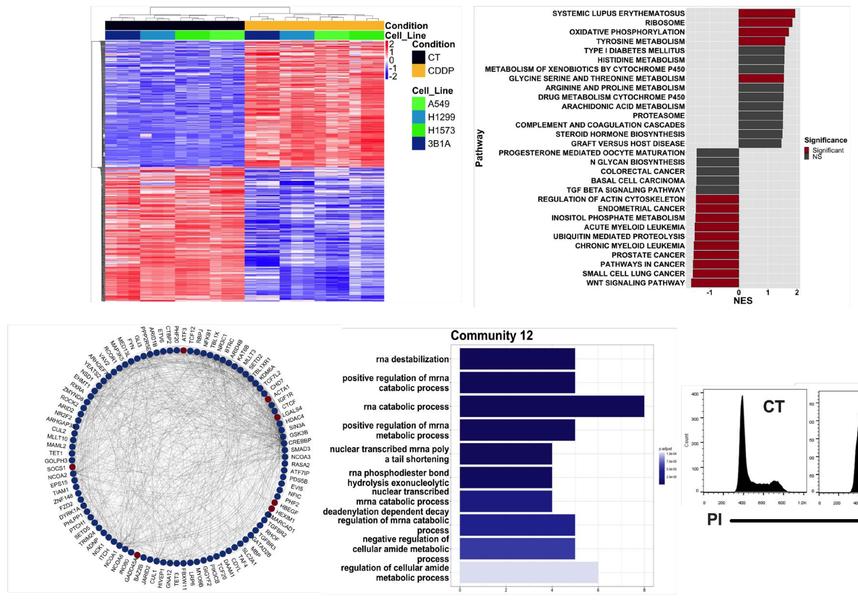
- CDDP induces cytotoxicity in the lung adenocarcinoma cell lines



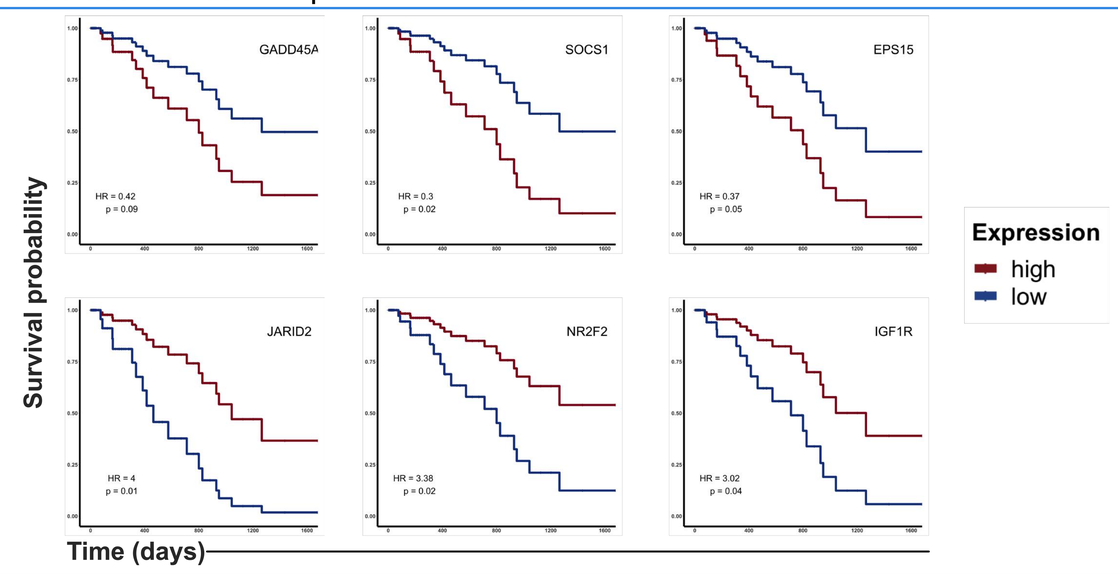




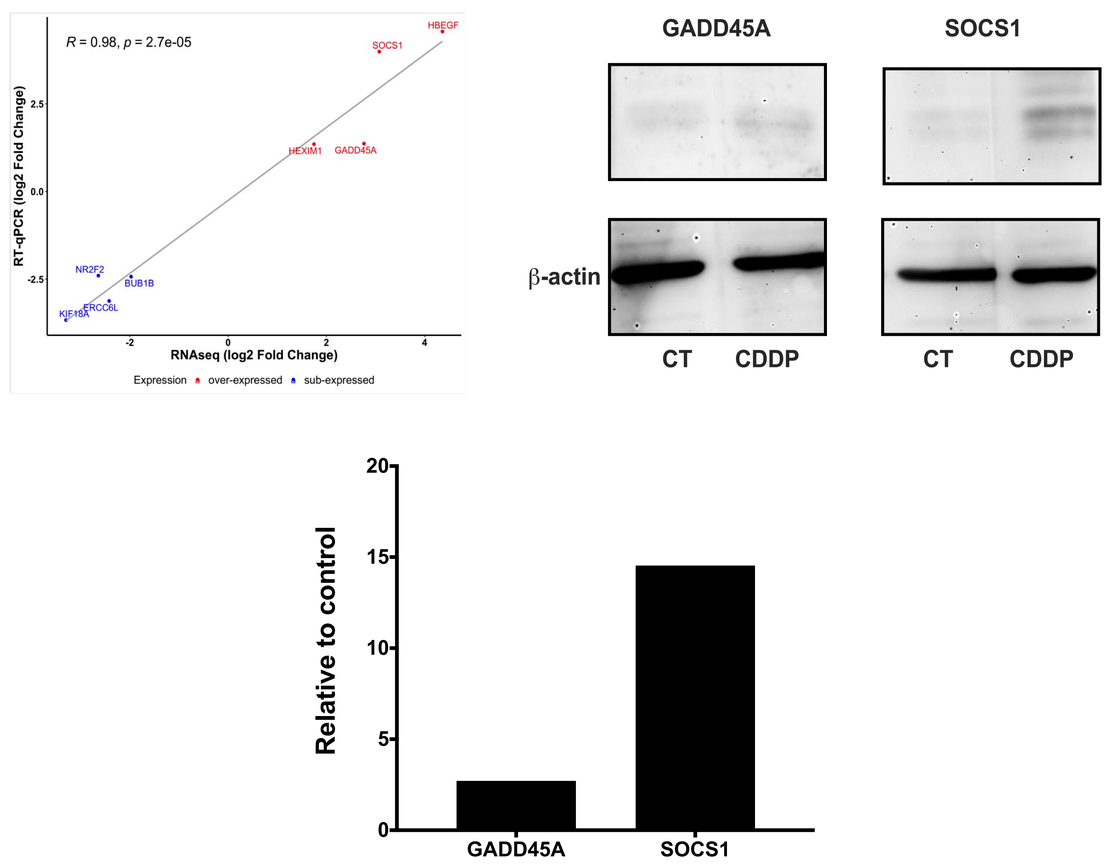
- Early CDDP-drug-tolerant persister cells reprogram their transcriptome to survive the cytotoxic effect of CDDP



- A gene expression signature related with early CDDP-drug-tolerant persister cells associates with patient survival



- **GADD45A** and **SOCS1** participate in the phenomenon of CDDP intrinsic resistance



5. CONCLUSIONS

- ✓ Lung adenocarcinoma cell lines exhibited distinct degrees of sensitivity to CDDP cytotoxic effect
- ✓ CDDP-drug-tolerant persister cells alter pathways to neutralize the drug and to decrease proliferation
- ✓ **Identified hub genes in CDDP-drug-tolerant persister cells could be employed as prognostic biomarkers or potential therapeutic targets**

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CONTACT

rodolfo_chvz@outlook.com

necrosnake91

@Rodolfo_chvz

SCAN ME